# Tutorial 8
# Deflection of beams

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1. The beam is made of two rods and is subjected to the concentrated load $P$ (see Fig. 1). Determine the maximum deflection of the beam if the moments of inertia of the rods are $I_{AB}$ and $I_{BC}$, and the modulus of elasticity is $E$.

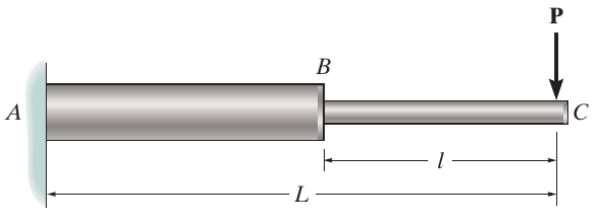



**Fig. 1**

2. Derive the equations of the deflection curve for a simple beam $AB$ loaded by a couple $M_0$ acting at distance $a$ from the left-hand support (see Fig. 2). Also, determine the deflection $\delta_0$ at the point where the load is applied.

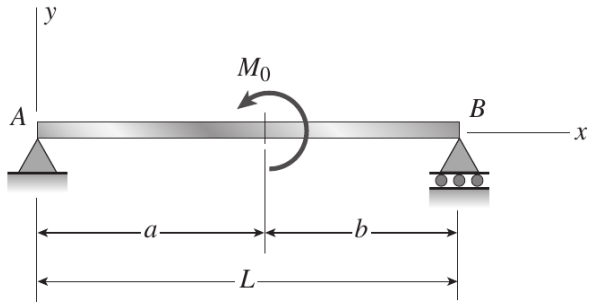



**Fig. 2**

3. Determine the maximum deflection of the simply supported beam (see Fig. 3) using singularity function. $E = 200\,\text{GPa}$ and $I = 65.0(10^6)\,\text{mm}^6$.

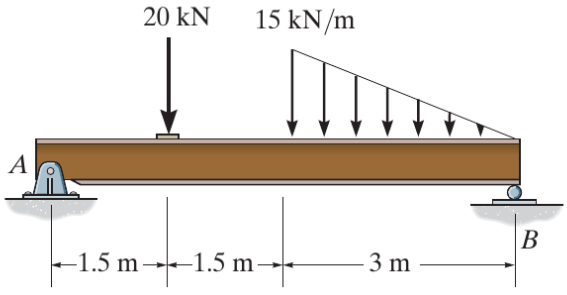



**Fig. 3**

4. Determine the slope and deflection of end $A$ of the cantilever beam shown in Fig. 4 using singularity function. $E = 200$ GPa and $I = 121(10^{-6})$ m$^4$.

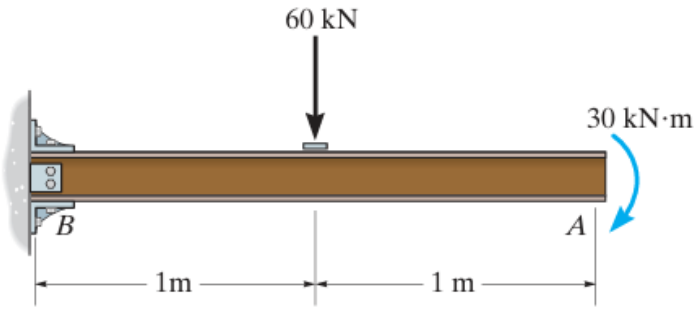



**Fig. 4**

5. The cantilever beam shown in the Fig. 5 supports a triangularly distributed load of maximum intensity $q_0$. Determine the deflection $\delta_B$ at the free end $B$. (Obtain the solution by determining the strain energy of the beam and then using Castigliano's theorem.)

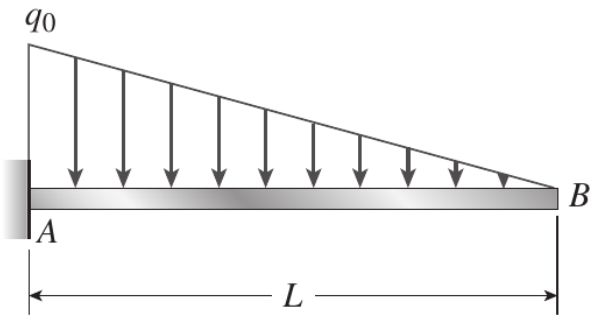



**Fig. 5**

6. Before the uniform distributed load is applied on the beam, there is a small gap of 0.2 mm between the beam and the post at $B$ (see Fig. 6). Determine the support reactions at $A$, $B$, and $C$. The post at $B$ has a diameter of 40 mm, and the moment of inertia of the beam is $I = 875(10^6)$ mm$^4$. The post and the beam are made of material having a modulus of elasticity of $E = 200$ GPa.

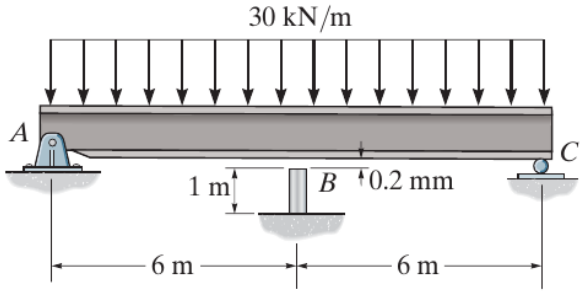



**Fig. 6**

7. Determine the reactions at the supports $A$, $B$, and $C$ using Castigliano's theorem, then draw the shear and moment diagrams. $EI$ is constant. (See Fig. 7)

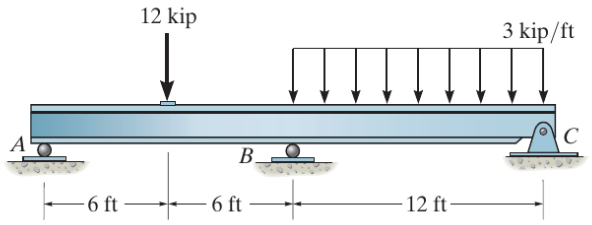



**Fig. 7**

8. The frame $ABC$ supports a concentrated load $P$ at point $C$ (see Fig. 8 ). Members $AB$ and $BC$ have lengths $h$ and $b$, respectively. Determine the vertical deflection $\delta_C$ and angle of rotation $\theta_C$ at end $C$ of the frame. Obtain the solution by using the modified form of Castigliano's theorem.

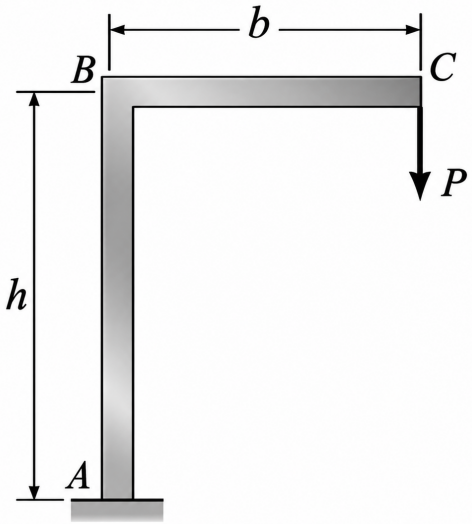



**Fig. 8**